WAPH-Web Application Programming and Hacking

Instructor: Dr. Phu Phung

Team 10 Project

Project Topic/Title - Mini Facebook Web App

Team members

1. Member 1 - Corey Brunner, email: brunnecj@mail.uc.edu
2. Member 2 - Mike Wojtkiewicz, email: wojtkiml@mail.uc.edu
3. Member 3 - Bashar Rasras, email: rasrasbr@mail.uc.edu
4. Member 4 - Pawan Poudyel, email: poudyepn@mail.uc.edu

Project Management Information

Source code repository (private access): https://github.com/waph-uc-sm26-team10/waph-teamproject

Project homepage (public): homepage

Revision History

Date	Version	Description
07/09/2026	0.0	README Implementation
07/18/2026	0.0	Add SSL and Database setup

Overview

The goal of this team project is to build a mini facebook application that takes into account security and full-stack web development.

Describe the overview of the project with a high-level architecture figure. *Note: You need to update this section according to each sprint*

System Analysis

(Start from Sprint 0, keep updating)

High-level Requirements

Users can log in to a system, change passwords, edit their profile credentials, make and edit those posts, and view comments

There should be super users that can access the database directly, disable/enable and delete users.

Logged in users can have real time chat messages with other logged in users.

System Design

(Start from Sprint 1, keep updating)

Use-Case Realization

Database

User Interface

Implementation

Local Setup

Secrets are kept out of the repo via a git-ignored `.env` file. Copy the template and fill in real values:

```
cp .env.example .env   # then edit .env with your credentials
```

Database - run the setup script (reads `.env`, creates the DB/user, and seeds the admin account):

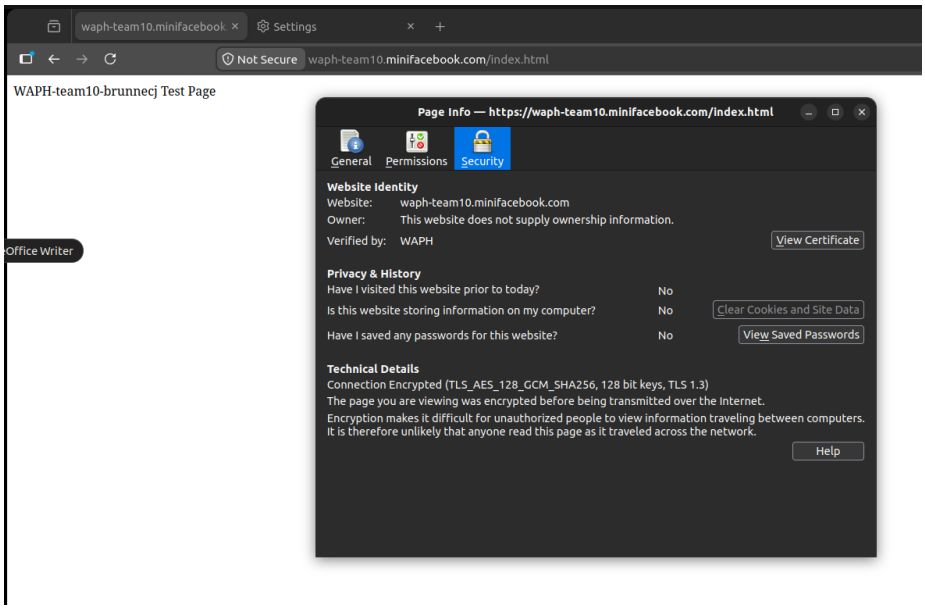
```
./database/setup.sh
```

SSL - generate a self-signed cert/key and install it (from the `sslkey/` folder):

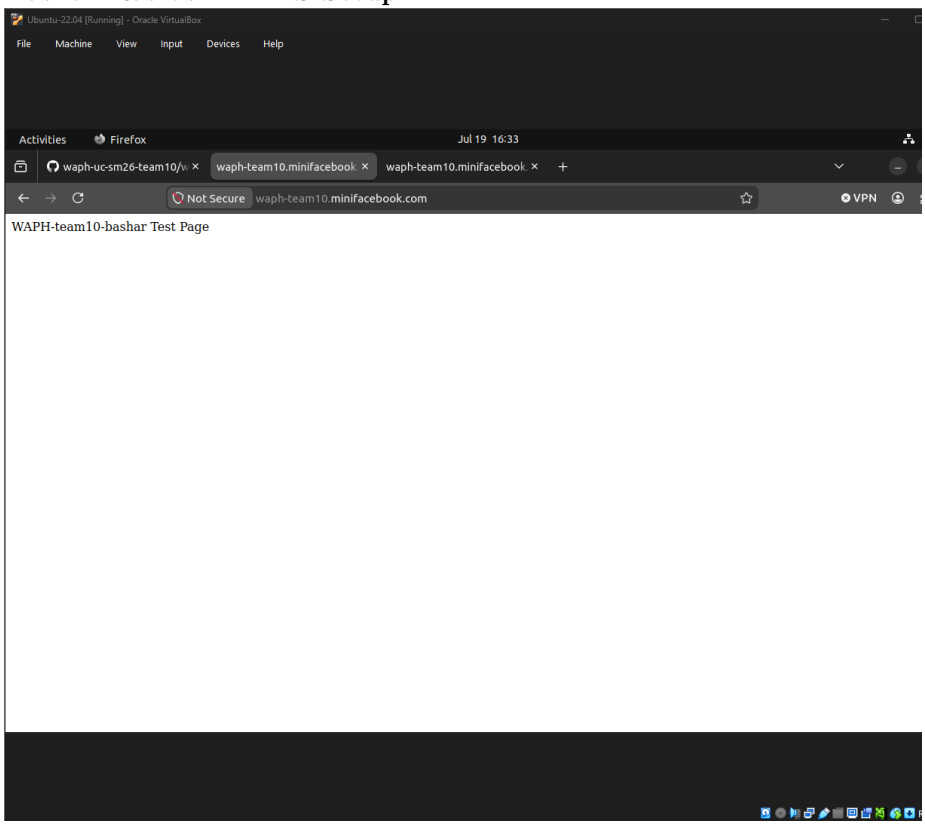
```
cd sslkey && ./ssl-keygen.sh
```

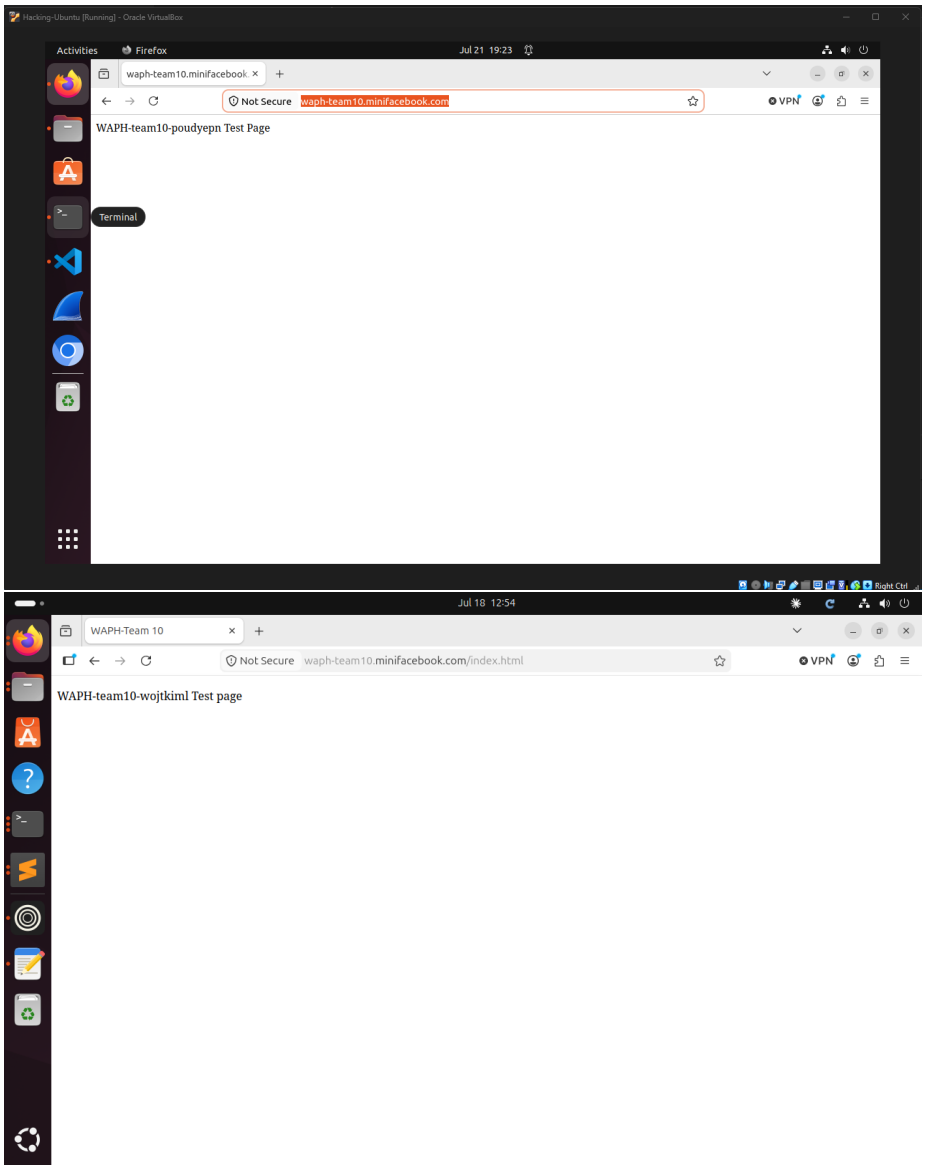
Local Domain Setup For Each Member

Corey Brunner HTTPS Setup

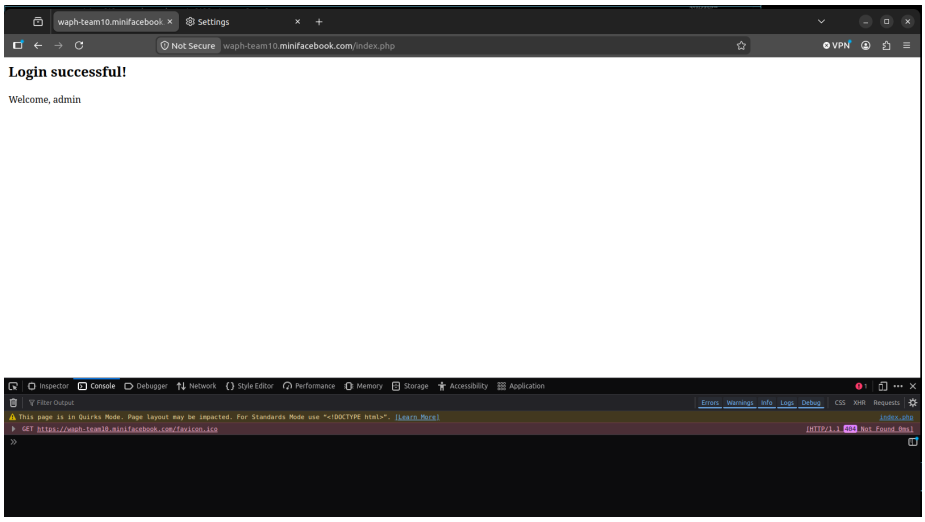


Bashar Rasras HTTPS Setup





Successful Local Domain Login Attempt



(Start from Sprint 1, keep updating. However, it is important to prepare the technology from Sprint 0)

For each new sprint cycle, update the implementation of your system (break it down into subsections). Please include some code snippets to illustrate the implementation.

Specify your team's development approach, including programming languages, databases, development, testing, and deployment environments.

Security analysis

Include a brief explanation of your implementation and the security aspects based on the following questions:

- How did you apply the security programming principles in your project?
- What database security principles have you used in your project?
- Is your code robust and defensive? How?
- How did you defend your code against known attacks such as XSS, SQL Injection, CSRF, Session Hijacking
- How do you separate the roles of super users and regular users?

Demo (screenshots)

You need to capture screenshots to demonstrate how your web application works. The screenshots must be accompanied by a short description of its functionalities following the implementation as below:

- Everyone can register a new account and then login

- Superuser can disable an account
  - The disabled account cannot log in
  - Superuser can enable the disabled account
  - The enabled user can log in
- A regular logged-in user can delete her own existing posts but cannot delete the posts of others
- CSRF attack to delete a post should be detected and prevented
- A regular logged-in user cannot access the link for superusers
- A logged-in user can have a real-time chat with other logged-in users

Software Process Management

(Start from Sprint 0, keep updating)

Introduce how your team uses a software management process, e.g., Scrum, and how your teamwork collaborates.

Scrum process

Sprint 0

Duration: DD/MM/YYYY-DD/MM/YYYY

Completed Tasks:

1. Task 1
2. Task 2
3. ...

Contributions:

1. Member 1, x commits, y hours, contributed in xxx
2. Member 2, x commits, y hours, contributed in xxx
3. Member 3, x commits, y hours, contributed in xxx
4. Member 4, x commits, y hours, contributed in xxx

Sprint 1

Duration: 07/14/2026-

Completed Tasks:

1. task 1 - Updated readme
2. task 2 - Upload demo screenshot
3. ...

Contributions:

1. Member 1, x commits, y hours, contributed in xxx
2. Member 2, x commits, y hours, contributed in xxx
3. Member 3, x commits, y hours, contributed in xxx
4. Member 4 - Pawan Poudyel, 6 commits, y hours, contributed in task 1, 2

Sprint Retrospection: _(Introduction to Sprint Retrospection:

Working through the sprints is a continuous improvement process. Discussing the completed sprint can improve the next sprint walk through a much more efficient one. Sprint retrospection is done once a sprint is finished and the team is ready to start another sprint planning meeting. This discussion can take up to 1 hour depending on the ideal team size of 4 members. Discussing good things that happened during the sprint can improve the team's morale, good team collaboration, appreciating someone who did a fantastic job solving a blocker issue, work well-organized, and helping someone in need. This will improve the team's confidence and keep them motivated. As a team, we can discuss what has gone wrong during the sprint and come up with improvement points for the next sprints. Few points can be like, need to manage time well, need to prioritize the tasks properly and finish a task in time, incorrect design lead to multiple reviews and that wasted time during the sprint, team meetings were too long which consumed most of the effective work hours. We can mention every problem is in the sprint which is hindering the progress. Finally, this meeting should improve your next sprint drastically and understand the team dynamics well. Mention the bullet points and discuss how to solve it.)

Good	Could have been better	How to improve?
	communication and meeting times	Have a set time where everyone gets in a call to discuess who is working on what

Sprint x

Duration: MM/DD/YYYY-MM/DD/YYYY

Completed Tasks:

1. Task 1
2. Task 2
3. ...
4.

Contributions:

1. Member 1, x commits, y hours, contributed in xxx
2. Member 2, x commits, y hours, contributed in xxx
3. Member 3, x commits, y hours, contributed in xxx
4. Member 4, x commits, y hours, contributed in xxx

Sprint Retrospection:

Good	Could have been better	How to improve?

Appendix

Include the content (in text, not as images) of the SQL files and all source code of your PHP files (with the file name).